(a) Semantic Granularity Learning

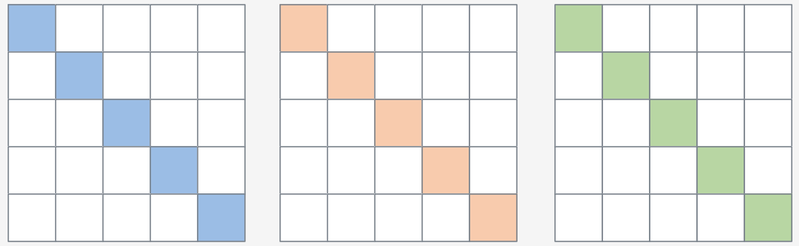
Signal
Signal Encoder
E^s
CLS token
SGL Adapter
Linear
Z^s

Image
Image Encoder
E^i
CLS token
SGL Adapter
Linear
Z^i

Clinical text
Text Encoder
E^t
CLS token
SGL Adapter
Linear
Z^t

Three bidirectional pairs

S ↔ T | I ↔ T | S ↔ I



B × B | B × B | B × B

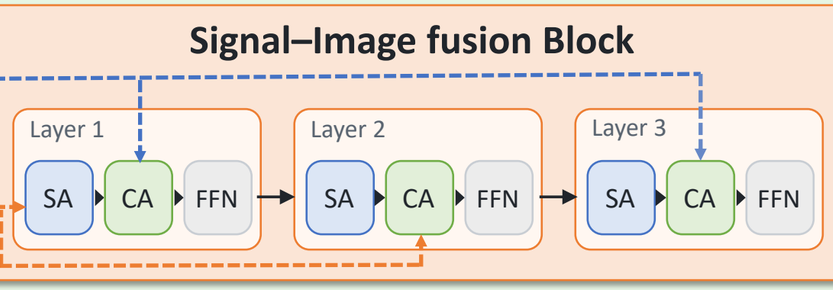
Diagonal cells: matched positives

Signal-Image-Text Joint Representation Space

(b) Concept Granularity Learning

E^s → CGL Adapter
E^i → CGL Adapter
Positive & hard-negative pairs

Signal–Image fusion Block



F^{si}
Matching head

Pair Results

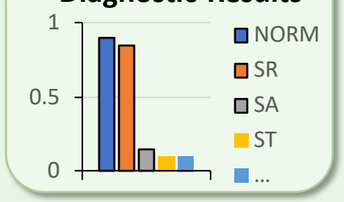
S_{pos} — I_{pos} ✓
S_{pos} — I_{neg} ×
S_{neg} — I_{pos} ×

Unified representation F^{si}

Concept-specific queries read disease evidence via the same fused tokens.

169 Disease Query Bank
Query Encoder
Disease query embeddings
Query-guided Tracing head
Multi-label classifier

Diagnostic Results



(c) Token Granularity Learning

Generate coherent and faithful clinical reports via the same fused tokens.

Unified representation F^{si}

TGL Adapter

Teacher forcing:
previous clinical tokens
[CLS] report: R [SEP] explanation: E [SEP]

Token + position embedding

Autoregressive decoder

LM head & softmax

Report R [SEP]
Explanation E [SEP]